

Where Can the Reserve Go?

Critical Tails, Causal Couplings, and the
Logarithmic Zero-Repair Problem

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Abstract

An exact causal realization must do more than reproduce one observed sequence: its continuation laws must remain compatible after every possible history. This note studies the cost of that compatibility in a small reserve model. Fresh increments have law Q , outputs must be iid with law P , and a nonnegative reserve absorbs their difference without repair. For a pair whose generating functions differ by $(1 - z)^4/64$, the established bounds are $\Omega(\log n) \leq \mathcal{B}_n \leq O(n^{1/4})$. We explain the origin of the question, give a self-contained account of these analytic bounds, and separate unrestricted statements from obstructions to particular representations. A review of recent causal transport, online discrepancy, randomness recycling, and finitary coding identifies several relevant mechanisms. Classical perpetuity theory supplies an explicit auxiliary process for the same pair with an exact critical $1/j$ tail. The missing result is a realization theorem connecting such mechanisms to the positive convolution equations of the reserve. The logarithmic upper bound remains open; the note makes no claim to improve the established asymptotic interval.

1 The cost of keeping alternatives compatible

Imagine a device that receives one small packet at a time and must immediately produce an output packet. The input and output have the same average size. A reserve can cover a temporary shortfall, but borrowing is forbidden. The output must also look exactly like a sequence of independent samples from a prescribed distribution. How much reserve must be present before the first input arrives?

Equal averages do not settle this question. A rule that balances today's packet can reveal information about tomorrow's output. A rule that preserves tomorrow's output law can leave a deficit today. The difficulty lies in satisfying both requirements through every possible continuation. Only one history is observed, but the rule must have a valid answer after each history that could have occurred.

The motivating manuscript *Who Pays the Bill?*, version 4, asks a related resource question for controlled stochastic channels [15]. It distinguishes a table whose time/action coordinates retain the same identity across histories from a response tree whose continuation may depend on the history. Its two-action construction places correlations between blocks that no admissible finite-horizon path can jointly expose. The manuscript thereby establishes, within its stated entropy contract, that allowing history-dependent continuations can remove an asymptotic surcharge. That is a concrete example of a resource saving hidden from every individual realized path.

The reserve problem isolates a different part of this phenomenon. Here the resource is an expected initial nonnegative integer, and each step obeys an additive conservation law. The two distributions below describe incoming and outgoing packet sizes; they are not the equal-entropy three-outcome action pair of the motivating manuscript. No identification of their resource measures is assumed. What carries over is the question about representation: which dependencies are compatible with exact observations, and which costs are artifacts of demanding more independence than the observations require?

1.1 A quartic mismatch that ordinary averages do not see

Our distributions are

$$\begin{aligned} P &= (1/8, 1/4, 1/4, 1/4, 1/8), \\ Q &= (7/64, 5/16, 5/32, 5/16, 7/64), \end{aligned} \tag{1}$$

on $\{0, 1, 2, 3, 4\}$. If $p(z) = G_P(z)$ and $q(z) = G_Q(z)$, then

$$p(z) - q(z) = \frac{(1-z)^4}{64}, \quad \mathbb{E}P = \mathbb{E}Q = 2. \tag{2}$$

The first three moments agree. The mismatch first appears at order four. A sufficiently broad reference law smooths it enough to permit an exact construction, but the known construction pays for that smoothing by increasing one common scale. Its reserve is $O(n^{1/4})$.

The same model forces a logarithmic lower bound. The gap between these estimates is the subject of this note. The fourth-root upper bound is the cost of a particular construction and its analysis. It is not a lower bound for all causal realizations, nor a proved optimum even within every possible common-scale class.

1.2 What the fan reveals

A useful observation is that conditional reserve means average exactly across the next generation. Large conditional reserves can therefore coexist with a modest average if they occur on histories of sufficiently small probability. But naming a branch “rare” is not a construction: its probability, its conditional reserve law, and every subsequent branch must fit one exact system of equations.

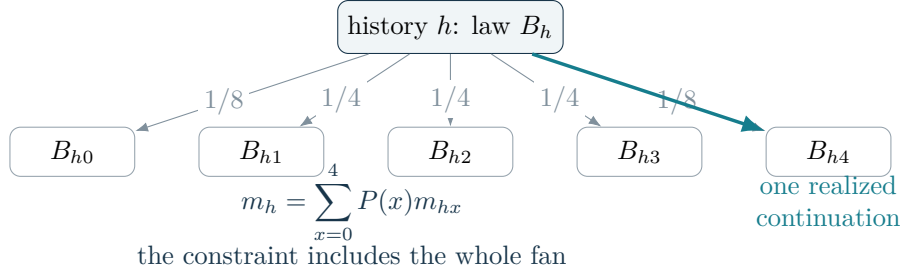


Figure 1: A schematic of the exact mean identity. The highlighted branch is illustrative; no zero-defect contact or optimal reserve value is assumed.

The finite optimizers initially suggested a simpler picture: a distinguished continuation could stay on the optimal frontier while its siblings carried the excess. That contact occurs at depths three and four, but fails at depth five in the supplied exact certificate calculations [5]. The useful lesson survives the failed extrapolation. Cost can be redistributed across continuations, but a single recursively optimal path does not describe the general mechanism.

There is consequently a precise distinction behind the intuition that “the mechanism is there.” Exact all-horizon reserve constructions, local redistribution, and critical-tail auxiliary dynamics do exist. Whether these ingredients admit an exact logarithmically efficient realization is the unresolved mathematical assertion. This note treats that assertion as a question, without presupposing its answer.

2 The exact model and the established interval

Write $\mathcal{H} = \{0, 1, 2, 3, 4\}^{<\infty}$ for finite output histories, including $\emptyset$. For $h = (x_1, \dots, x_t)$, set $|h| = t$, $S(h) = \sum_i x_i$, and $P(h) = \prod_i P(x_i)$. All logarithms are natural unless an entropy is explicitly measured in bits.

Definition 2.1 (Exact zero-repair tree). A depth- n tree is a family of probability laws B_h on $\mathbb{Z}_{\geq 0}$, $|h| \leq n$, such that

$$Q * B_h = \sum_{x=0}^4 P(x) \delta_x * B_{hx}, \quad |h| < n. \quad (3)$$

Let $\mathcal{K}_n$ denote its feasible root laws and define

$$\mathcal{B}_n = \inf_{B \in \mathcal{K}_n} \mathbb{E}B, \quad \mathcal{B}_0 = 0. \quad (4)$$

There is no terminal condition requiring the reserve to vanish. An infinite tree satisfies (3) at every finite history; its laws must be proper probabilities, but may have infinite means.

2.1 What causality requires

The operational interpretation uses a reserve $J_t \geq 0$, fresh iid inputs $Y_t \sim Q$, and outputs X_t with joint law $P^{\otimes n}$:

$$J_t = J_0 + \sum_{i=1}^t (Y_i - X_i). \quad (5)$$

The initial reserve and any initial randomization are independent of future inputs. At time t , the output may use the current input, the past, the reserve, and independent auxiliary randomness, but no future input. In particular,

$$\text{Law}(X_{t+1} \mid X_1, \dots, X_t) = P. \quad (6)$$

The stronger condition obtained by conditioning also on the entire hidden state and input past is not imposed. Confusing these filtrations would restrict the admissible class.

Lemma 2.2 (Causal lift). *Every exact tree admits the operational realization above, with $B_h = \text{Law}(J_t \mid X_{1:t} = h)$. This includes infinite trees with infinite reserve means.*

Proof. Draw $J_0 \sim B_\emptyset$. At history h , draw a fresh $Y \sim Q$ and put $s = J_t + Y$. Select $X = x$ with probability

$$K_h(x \mid s) = \frac{P(x)B_{hx}(s-x)}{(Q * B_h)(s)}, \quad (7)$$

where masses at negative indices are zero. Equation (3) makes this a probability kernel whenever its denominator is positive; zero-denominator states have zero probability and may be assigned arbitrarily. Put $J_{t+1} = s - X$. Conditional on h , the joint law of (X, J_{t+1}) is $P(x)B_{hx}(j)$. This proves nonnegativity, (6), and the induction on conditional reserve laws. The countable sequence of kernels also defines an infinite process. No moment calculation is used. $\square$

Conversely, an operational realization gives (3): conditional on h , the left side is the law of $J_t + Y_{t+1}$ and the right side is the law of $X_{t+1} + J_{t+1}$. Thus the tree equation preserves the full intended causal freedom.

Lemma 2.3 (Root truncation). *For an exact depth- n tree, its operational realization can be changed to one whose root is $\min(J_0, 4n)$, without changing any output or input. Consequently the infimum in (4) is attained by a finite-support tree. An infinite tree with*

$$\mathbb{P}(J_0 \geq j) \leq C/j \quad (j \geq 1) \quad (8)$$

would imply $\mathcal{B}_n \leq CH_{4n} = O(\log(n+1))$.

Proof. Set $D = (J_0 - 4n)_+$ and $J'_t = J_t - D$ for $0 \leq t \leq n$. If $D > 0$, then $J_t \geq J_0 - 4t$, so $J'_t \geq 4n - 4t \geq 0$; if $D = 0$, nothing changes. The same causal algorithm can retain D as auxiliary initial state. The inputs remain fresh and the unconditional output law remains unchanged. We do not condition on D and assert a new iid output law.

The root is now supported on $\{0, \dots, 4n\}$ and every later reserve is at most $8n$. Conditional masses therefore lie in a finite product of simplexes with closed linear constraints. Nonemptiness follows, for example, by starting at $4n$ and sampling independent outputs. Compactness gives attainment. Finally,

$$\mathbb{E} \min(J_0, 4n) = \sum_{j=1}^{4n} \mathbb{P}(J_0 \geq j) \leq C \sum_{j=1}^{4n} \frac{1}{j}.$$

□

Theorem 2.4 (Established all-horizon estimates). *For (1),*

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \kappa_0 := \frac{\log(594184/543559)}{\log 16 \log(524288/74273)} \approx 0.0164347253, \quad (9)$$

$$\mathcal{B}_n \leq 4(4096 + 2465n)^{1/4} \leq 36n^{1/4} \quad (n \geq 1). \quad (10)$$

These are the analytic baseline results of the project, not conclusions drawn from the literature review. Complete proofs are included in Appendices A and B. The finite-horizon contact statements are reported from companion certificates [5] and are not needed for either bound.

Problem 2.5 (Unrestricted logarithmic reserve). Determine whether

$$\sup_{n \geq 2} \frac{\mathcal{B}_n}{\log n} < \infty. \quad (11)$$

Equivalently, does a finite C satisfy $\mathcal{B}_n \leq C \log(n+1)$ for every $n \geq 1$? A positive answer, together with (9), gives $\mathcal{B}_n = \Theta(\log n)$. A negative answer means that the displayed ratio is unbounded; it need not tend to infinity.

The critical infinite-tree route of Lemma 2.3 is sufficient for a positive answer. Its necessity has not been established. Finite-horizon optimizers may depend on n without admitting a common infinite extension of the proposed form.

3 What the current invariants actually say

3.1 The fan identity and the scope of contact

For a finite-mean exact tree put $m_h = \mathbb{E}B_h$. Equal input and output means give

$$m_h = \sum_x P(x) m_{hx}. \quad (12)$$

At a node with $r = n - |h|$ steps remaining define $\Delta_h = m_h - \mathcal{B}_r \geq 0$. Iterating (12) proves

$$\sum_{|g|=s} P(g) \Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}, \quad 0 \leq s \leq r. \quad (13)$$

This is an accounting identity for the full fan of continuations. It does not identify a unique branch that pays the increment. In the supplied optimal-face calculations, $\Delta_{(4)} = 0$ for depth three and $\Delta_{(4)} = \Delta_{(4,4)} = 0$ for depth four. Depth five instead forces $\Delta_{(4)} > 0$ at every optimum [5]. The unconditional identity (13) survives all these changes.

3.2 Two averages that a small-reserve construction must reconcile

For $0 < z < 1$ define

$$P_z(x) = \frac{P(x)z^x}{p(z)}, \quad c(z) = \frac{\log(p(z)/q(z))}{-\log z} > 0. \quad (14)$$

If $F_h(z) = \mathbb{E}_{B_h} z^J$, iteration of (3) yields

$$\mathbb{E}_{P_z^{\otimes t}} F_h(z) = \left(\frac{q(z)}{p(z)} \right)^t F_{\emptyset}(z), \quad |h| = t. \quad (15)$$

Twice applying Jensen's inequality, for finite reserve means, gives

$$\mathbb{E}_{P_z^{\otimes t}} m_h \geq tc(z) + \frac{-\log F_{\emptyset}(z)}{-\log z} \geq tc(z). \quad (16)$$

Thus a proposed logarithmic depth- n construction must reconcile

$$\mathbb{E}_{P^{\otimes n}} m_h = O(\log n), \quad \mathbb{E}_{P_z^{\otimes n}} m_h \geq nc(z). \quad (17)$$

These are statements about weighted averages, not a pointwise assertion on every tilted-typical history. In particular, repeatedly keeping the same conditional mean in every child cannot give logarithmic cost.

The density ratio

$$\Lambda_h(z) = \frac{P_z^{\otimes t}(h)}{P(h)} = \frac{z^{S(h)}}{p(z)^t}$$

is a positive mean-one martingale in the output filtration. It quantifies the change in weights in (17). Its harmonicity does not prove that the whole reserve law can be indexed by Λ_h .

3.3 All local freedom is expressible by signed currents

Proposition 3.1 (Exact current parametrization). *Let F be a probability generating function. Put $D(z) = (1-z)^3 F(z)$, $d_0 = -D/64$, $d_1 = D/64$, and $d_2 = d_3 = d_4 = 0$. All exact one-step families are parametrized by*

$$P(x)(F_x - F) = d_x + zH_x - H_{x-1}, \quad 0 \leq x \leq 4, \quad (18)$$

where $H_{-1} = H_4 = 0$ and $H_0, \dots, H_3$ are generating functions of signed ℓ^1 sequences with $H_i(1) = 0$. For chosen currents the resulting children are probability laws if and only if every coefficient of every F_x is nonnegative.

Proof. Evaluation at $z = 1$ gives child mass one. Multiplication by z^x and summation cancels the currents and gives

$$\sum_x P(x) z^x F_x = p(z) F(z) - \frac{(1-z)^4}{64} F(z) = q(z) F(z).$$

Conversely, for an exact family put $A_x = P(x)(F_x - F) - d_x$. Then $\sum_x z^x A_x = 0$ and $A_x(1) = 0$. Define

$$H_3 = -A_4, \quad H_2 = zH_3 - A_3, \quad H_1 = zH_2 - A_2, \quad H_0 = zH_1 - A_1.$$

The remaining equation is $A_0 = zH_0$. These currents lie in ℓ^1 and have zero total mass. $\square$

When the first absolute moments exist, differentiation gives

$$P(x)(m_x - m) = H'_x(1) - H'_{x-1}(1). \quad (19)$$

For an infinite tree with critical tails, this is the wrong integrability assumption. Instead, if a signed sequence satisfies $a(j) = Aj^{-2} + o(j^{-2})$, write $\tau(a) = A$. Finite shifts preserve τ , and $\tau((1-z)^3F) = 0$ whenever $\tau(F)$ exists. Under these coefficient asymptotics, (18) becomes

$$P(x)\{\tau(F_x) - \tau(F)\} = \tau(H_x) - \tau(H_{x-1}). \quad (20)$$

For a probability mass function with this asymptotic, $\mathbb{P}(J \geq j) \sim A/j$ when $A > 0$. Finite total variation alone does not guarantee the existence of τ .

The current equations solve exactness and normalization algebraically. Their remaining content is global positivity: the child laws obtained now must admit positive continuations at all later depths. Matching a mean or a tail coefficient is not a substitute for that requirement.

3.4 Order information cannot be discarded wholesale

Proposition 3.2 (Permutation-invariant histories are insufficient). *There is no exact infinite tree of proper probability laws for (1) in which permuting a history leaves its reserve law unchanged. Infinite reserve means do not avoid this obstruction. Every finite depth- n tree with this invariance and finite means satisfies*

$$\mathbb{E}B_\emptyset \geq \frac{\gamma}{8}n, \quad \gamma = \frac{\log(122725639/134217728)}{\log(1/64)} > 0. \quad (21)$$

The proof is in Appendix C. In particular, an infinite family $B_h = B_{\Lambda_h(z_*)}$ cannot work: that score forgets order. This is an obstruction to a representation, not a stronger lower bound for unrestricted trees. A harmonic score may still describe a tail amplitude while other state coordinates encode order.

4 Critical tails as an exact auxiliary mechanism

The equation $\mathbb{E}A^\alpha = 1$ is the tail-index condition in Kesten–Goldie theory for random affine recursions. Under contraction on the logarithmic scale and nonarithmeticity, a stationary perpetuity has a power-law tail [12, 13]. In the present pair the critical case $\alpha = 1$ can be realized explicitly.

Theorem 4.1 (A critical perpetuity from the quartic pair). *Let $X_1, X_2, \dots$ be iid with law P , and put $A_i = Q(X_i)/P(X_i)$. Define*

$$R = 1 + \sum_{k \geq 1} \prod_{i=1}^k A_i. \quad (22)$$

Then $R < \infty$ almost surely, $R \stackrel{d}{=} 1 + AR'$ for an independent copy R' , and

$$\mathbb{P}(R > j) \sim \frac{1}{d_* j}, \quad \mathbb{E} \min(R, M) \sim \frac{\log M}{d_*}, \quad (23)$$

where

$$\begin{aligned} d_* &= D(Q\|P) = \sum_x Q(x) \log \frac{Q(x)}{P(x)} \\ &= \frac{7}{32} \log \frac{7}{8} + \frac{5}{8} \log \frac{5}{4} + \frac{5}{32} \log \frac{5}{8} \approx 0.0368166604. \end{aligned} \quad (24)$$

This is an application of established perpetuity theory to the specified distributions.

Proof. The multiplier takes values $7/8, 5/4, 5/8$, and

$$\mathbb{E}A = 1, \quad \mathbb{E} \log A = -D(P\|Q) < 0, \quad \mathbb{E}(A \log A) = d_* > 0.$$

The strong law makes $\prod_{i=1}^k A_i$ decay exponentially almost surely, proving convergence of (22). Splitting off A_1 gives its distributional fixed-point equation.

All positive moments of A are finite. The law of $\log A$ is nonarithmetic: its support differences include $\log 2$ and $\log(7/5)$, and a rational relation between them contradicts unique prime factorization. There is no real r with $1 + Ar = r$ almost surely, because A is nonconstant. Thus the hypotheses of the Kesten–Goldie theorem hold with additive term 1 and exponent 1 [12, 13].

Goldie’s constant, also recorded in [6, Lemma 6.1, Eq. (6.3)], is

$$\frac{\mathbb{E}[(1 + AR') - AR']}{\mathbb{E}(A \log A)} = \frac{1}{d_*}.$$

The numerator is the expectation of a pointwise difference equal to one; no subtraction of two infinite expectations is made. Finally integrate the tail:

$$\mathbb{E} \min(R, M) = \int_0^M \mathbb{P}(R > u) du \sim d_*^{-1} \log M.$$

□

The recursion $R_t = 1 + A_t R_{t-1}$ is causal with respect to its iid multipliers. Starting from $R_0 = 0$, one has $\mathbb{E}R_t = t$ by induction, while R_t converges in distribution to the proper law in (22). The stationary mean is infinite. This gives an elementary example of a tight family whose means grow through rare large values, precisely the kind of nonuniform integrability that finite-mean intuition can miss.

It also retains order. For $T_a(r) = 1 + ar$,

$$T_a(T_b(r)) - T_b(T_a(r)) = a - b.$$

The product likelihood ratio alone does not determine this state. These are genuine structural advantages over a law indexed only by a commutative score.

The missing transfer. Theorem 4.1 does not construct a reserve tree. Its increments $1 + (A_t - 1)R_{t-1}$ are unbounded, whereas $Y_t - X_t \in [-4, 4]$. Rounding R gives an integer law with a critical tail but supplies no children satisfying (3). A bridge must preserve the additive balance, fresh-input causality, exact iid outputs, and nonnegativity simultaneously.

4.1 Criticality does not control slowly varying losses

Coupled affine recursions can have tails more delicate than their scalar components. Damek and Matsui show that triangular systems with matching critical indices may acquire powers of $\log j$ in addition to the power-law factor [6]. This matters because, for $\beta \geq 0$,

$$\mathbb{P}(J \geq j) \asymp \frac{(\log j)^\beta}{j} \implies \mathbb{E} \min(J, M) \asymp (\log M)^{\beta+1}. \quad (25)$$

Equation (25) follows by summation or integration. It is not asserted as the tail of the reserve model. It shows why preserving the exponent 1 in an auxiliary construction is weaker than establishing the uniform C/j estimate needed by Lemma 2.3.

5 What the surrounding literature supplies

The following is a targeted literature map, with versions checked through 5 September 2026. References to recent preprints identify their claims and assumptions; they are not presented as independently certified solutions of the reserve problem. Only the perpetuity application in Section 4 is used as an external probabilistic theorem in a proof here.

5.1 Causal transport: preserve the filtration as well as the law

Kršek and Pammer establish general duality and dual attainment for causal and bicausal transport [14]. Their Theorem 2.1 allows measurable costs bounded above, and the general version allows an integrable marginal upper bound. The dual variables incorporate the temporal conditional-independence restrictions through martingale conditions. This provides a substantially broader class of potentials than a fixed collection of moment tests.

The relevant direction for (3) is one-sided causality from the augmented source $(J_0, Y_{1:n})$ to $X_{1:n}$. Requiring causality in the reverse direction changes the problem. Moreover, a theorem for fixed marginals does not itself perform the outer optimization over $\text{Law}(J_0)$ or supply a horizon-uniform estimate. A $+\infty$ barrier cost for negative reserves also requires care when applying a theorem stated for upper-bounded costs. General duality is an exact structural tool, not the missing logarithmic estimate.

5.2 Strong approximation: a small timing allowance is a real allowance

Bartl, Beiglböck, Pammer, Schrott, and Zhang develop a continuous-time adapted Wasserstein framework that accommodates small shifts in information timing [3]. Their Proposition 7.3 constructs an ε -bicausal coupling of Brownian motion W and a scaled symmetric random walk W^n with

$$\mathbb{E}\|W - W^n\|_\infty \leq \frac{C \log n}{\sqrt{n\varepsilon}}. \quad (26)$$

The construction uses strong couplings within blocks and allows information tolerance across the block scale. This is useful evidence that information timing must be priced explicitly when transporting a strong approximation into an online model.

For our question, small discrepancy on a jointly sampled full pair of paths is insufficient. The initial reserve must not encode future Q -inputs, and negative reserves must be excluded almost surely rather than with high probability. Equation (26) does not remove either obligation, and gives no fourth-root lower bound for the one-sided discrete reserve problem.

5.3 Online discrepancy: invariant marginals with flexible joint couplings

Aden-Ali’s July 2026 preprint gives an explicit online signing algorithm in which

$$\sum_{i=1}^t \varepsilon_i v_i = G_{t,1} + G_{t,2} + G_{t,3}, \quad G_{t,j} \sim N(0, I), \quad (27)$$

for every fixed input vector sequence with $\|v_i\|_2 \leq 1$ [1]. Each auxiliary walk preserves its Gaussian marginal through a Metropolis transition. A coupling of the three increments makes their sum an allowed sign, although the auxiliaries need not be independent. The resulting prefix discrepancy has the optimal $O(\sqrt{\log T})$ order with high probability.

Altschuler and Tikhomirov use a related compactly supported Metropolis fixed-point walk for online Beck–Fiala discrepancy [2]. Their work makes explicit that invariant-law methods can be designed with a support boundary rather than only Gaussian tails.

The relevant common mechanism is to solve marginal invariance locally and use joint coupling freedom to impose a pathwise constraint. The additional reserve-model requirement is (6): the chosen outputs must have the prescribed iid law. Online discrepancy algorithms are allowed to choose temporally dependent signs. Therefore their signing guarantees do not directly give (3), a nonnegative reserve, or an initial-reserve estimate.

5.4 Randomness recycling: a logarithm in a different resource

Draper and Saad give exact online sampling by recycling unused randomness, with amortized entropy overhead ε and persistent space $O(\log(1/\varepsilon))$ [8]. Draper, Harris, and Saad’s July 2026 preprint treats real probabilities through rational approximations and obtains an expected entropy overhead $O(\log n)$ with $O(\log n)$ persistent bits [7]. A companion preprint proves space lower bounds for samplers approaching the entropy limit [9].

These results are especially close to the motivation: local state can preserve reusable information that would be wasted by independent one-shot implementations. But a bit-memory bound, a number of random bits consumed, and $\mathbb{E}J_0$ are different quantities. Our inputs supply exactly one bounded Q -increment per step; the sampler may consume a variable number of fair bits. An equivalence between these accounting models is not established.

5.5 Catalytic stochastic order: the independent auxiliary boundary

Fritz characterizes asymptotic and catalytic stochastic orders in generic cases through inequalities related to cumulant-generating functions [10]. In the catalytic formulation, one adds the same independent auxiliary variable to two laws and asks for stochastic dominance. His sufficient conditions use strict inequalities, including the mean comparison in the real-valued case.

Our equal-mean pair does not meet that strict regime. For a finite-mean independent auxiliary Z , stochastic dominance between $P * \text{Law}(Z)$ and $Q * \text{Law}(Z)$ would, by equality of means, force equality of these convolutions. Their generating functions then contradict (2). History-dependent reserve laws avoid the demand that the same independent auxiliary return unchanged; Fritz’s theorem does not quantify the cost of this freedom in the present model.

5.6 Finitary coding: moment restrictions depend on what is encoded

Gabor’s September 2025 preprint proves that an equal-entropy finitary map between iid $\mathbb{Z}^d$ processes with finite $d/2$ moment of its coding radius preserves informational variance. Exponentially decaying coding radius forces the marginals to agree up to symbol permutation [11]. Informational variance means $\text{Var}(-\log P(X))$, not $\text{Var}(X)$.

For the present pair, $H_2(P) = 2.25$ and $H_2(Q) \approx 2.16563479$ bits. The equal-entropy hypothesis is absent, auxiliary randomness is allowed, and the reserve need not be a finitary factor. Thus this result does not obstruct the unrestricted zero-repair construction.

Chazottes, Gallo, and Takahashi relate coding-volume moments to Gaussian concentration [4]. Their revised preprint treats left finitary coding, in which each output uses a finite random portion of the past; Theorem 4.7 gives a concentration estimate controlled by the first moment of that left coding radius. This is a useful way to distinguish random memory length from a fixed memory bound. It still imposes no additive conservation law, and a coding radius cannot be identified with the reserve without a separate argument.

6 A precise statement of the missing realization result

Several mechanisms are now visible: an exact local nullspace described by currents, causal auxiliary dynamics with critical tails, and joint couplings that preserve useful marginals while controlling sums. Their coexistence does not yet establish that they can be combined in the reserve model.

The unrestricted finite-horizon problem can be stated without committing to any auxiliary representation. Let μ be a probability on $\{0, \dots, 4n\}$, and let $\Pi_c(\mu \otimes Q^{\otimes n}, P^{\otimes n})$ be the couplings in which $X_{1:t}$, conditional on $(J_0, Y_{1:n})$, depends only on $(J_0, Y_{1:t})$. Define

$$V_n(\mu) = \min_{\pi \in \Pi_c} \pi \left\{ \min_{0 \leq t \leq n} \left(J_0 + \sum_{i=1}^t (Y_i - X_i) \right) < 0 \right\}.$$

All alphabets here are finite, so the causal constraints define a compact polytope and the minimum exists. Lemmas 2.2 and 2.3 give the exact equivalent formulation

$$\mathcal{B}_n = \min \left\{ \sum_{j=0}^{4n} j \mu(j) : V_n(\mu) = 0 \right\}. \quad (28)$$

Equation (28) is a reformulation, not an improved estimate. It records what any proposed use of transport, coding, or invariant processes must achieve: zero probability of failure under the correct causal coupling, with a small initial mean.

For the infinite-tree route, the sufficient statement remains

$$\left. \begin{array}{l} B_h \text{ proper probability laws on } \mathbb{Z}_{\geq 0}, \quad h \in \mathcal{H}, \\ Q * B_h = \sum_x P(x) \delta_x * B_{hx}, \\ \sup_{j \geq 1} j \sum_{k \geq j} B_{\emptyset}(k) < \infty \end{array} \right\} \implies \mathcal{B}_n = \Theta(\log n). \quad (29)$$

There is no condition here that the laws share a scale, depend on one likelihood score, have finite memory, or follow a contact path. Infinite means at individual nodes are allowed. Even the coefficient asymptotic $B_{\emptyset}(j) \sim A/j^2$ is stronger than necessary: the survival bound in (29) is enough.

Conversely, an exact infinite tree with root survival $o(1/j)$ is impossible. Lemma 2.3 would give $\mathcal{B}_n = o(\log n)$, since the truncated tail sum of an $o(1/j)$ sequence is $o(\log n)$, contradicting (9). Thus $1/j$ is a genuine critical boundary for this route, although a proper infinite tree at that boundary has not been constructed.

Mathematical status. The unrestricted interval remains

$$\Omega(\log n) \leq \mathcal{B}_n \leq O(n^{1/4}).$$

The current parametrization, the order obstruction, and the critical perpetuity describe structure without narrowing that interval. No result reviewed here proves (11) or its negation. The finite-horizon question and the stronger critical infinite-tree existence question must remain distinct.

The motivation is therefore sharper than a claim that a logarithm must be hiding in a fourth-root calculation. The known constructions demonstrate that exactness leaves substantial freedom in dependencies. The open issue is whether that freedom can maintain nonnegative continuation laws at the critical cost. A successful description must make that compatibility verifiable throughout the tree.

A Proof of the logarithmic lower bound

Fix $0 < z < 1$ and abbreviate $p = p(z)$, $q = q(z)$, $c = c(z)$. Take an optimal depth- r tree, an integer $1 \leq t \leq r$, and $b = \mathcal{B}_{r-t}$. Subtree feasibility gives $m_h - b \geq 0$ for $|h| = t$. The tilted bound (16) and

$$\frac{P_z^{\otimes t}(h)}{P(h)} = \frac{z^{S(h)}}{p^t} \leq p^{-t}$$

imply

$$tc \leq \mathbb{E}_{P_z^{\otimes t}} m_h \leq b + p^{-t}(\mathcal{B}_r - b).$$

Hence

$$\mathcal{B}_r \geq (1 - p^t)\mathcal{B}_{r-t} + tc p^t. \quad (30)$$

Iterating at multiples of t , and using monotonicity under pruning, yields

$$\mathcal{B}_n \geq tc \left[1 - (1 - p^t)^{\lfloor n/t \rfloor} \right], \quad 1 \leq t \leq n. \quad (31)$$

Choose

$$t = t_n = \left\lfloor \frac{\log n - 2 \log \log n}{-\log p} \right\rfloor.$$

For large n , $t_n > 0$, $t_n / \log n \rightarrow 1 / (-\log p)$, and $\lfloor n/t_n \rfloor p^{t_n} \rightarrow \infty$. The bracket in (31) tends to one. Therefore

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{\log(p(z)/q(z))}{(-\log z)(-\log p(z))}. \quad (32)$$

At $z = 1/16$,

$$p = \frac{74273}{524288}, \quad q = \frac{543559}{4194304},$$

which gives (9). In particular no exact infinite tree has a finite root mean.

B An explicit all-horizon fourth-root construction

For $a > 0$ let R_a be the sum of four independent geometric variables on $\mathbb{Z}_{\geq 0}$, each of mean a . Its mass and mean are

$$b_a(j) = \binom{j+3}{3} \frac{a^j}{(1+a)^{j+4}}, \quad \mathbb{E}R_a = 4a. \quad (33)$$

Let $F_a(j) = \mathbb{P}(R_a \leq j)$, and extend masses and CDFs by zero at negative indices. Set

$$a_0 = 8, \quad a_{r+1} = a_r + \frac{512}{a_r^3}. \quad (34)$$

We first prove $Q * R_{a_{r+1}} \geq_{\text{st}} P * R_{a_r}$.

B.1 The discrete defect estimate

For $\Delta f(j) = f(j) - f(j-1)$, (2) gives

$$\sum_x P(x) F_a(j-x) - \sum_y Q(y) F_a(j-y) = \frac{\Delta^3 b_a(j)}{64}. \quad (35)$$

Put $D_j = (j+1)(j+2)(j+3)$. Direct expansion yields

$$\frac{\Delta^3 b_a(j)}{b_a(j)} = \frac{N(a, j)}{a^3 D_j}, \quad N(a, j) = 6a^3 - 18ja^2 + 9j(j-1)a - j(j-1)(j-2).$$

The identity

$$N(a, j) + 25D_j = 6(a - 2j)^2(a + j) + 9j(j - 1)a + 153j^2 + 273j + 150$$

proves $\Delta^3 b_a(j) \geq -25a^{-3}b_a(j)$ for integers $j \geq 0$. If $j < a/4$, then

$$N(a, j) \geq 6a^3 - 18ja^2 - j^3 > a^3(6 - 18/4 - 1/64) > 0.$$

Thus a negative defect can occur only for $j \geq a/4$.

B.2 Scale growth dominates the negative defect

The negative-binomial CDF satisfies

$$-\partial_u F_u(j) = \frac{j+4}{1+u} b_u(j). \quad (36)$$

Also $\partial_u \log b_u(j) \geq -4/(1+u)$. For $a \leq u \leq 2a$ this gives $b_u(j) \geq b_a(j)/16$, and for $j \geq a/4$ one has $(j+4)/(1+u) \geq 1/8$.

Take $a \geq 8$, $\delta = 512/a^3$, and $a' = a + \delta \leq 2a$. By monotonicity of each CDF term and $Q(0) = 7/64$,

$$\begin{aligned} \sum_y Q(y) \{F_a(j-y) - F_{a'}(j-y)\} &\geq \frac{7}{64} \int_a^{a'} \frac{j+4}{1+u} b_u(j) du \\ &\geq \frac{7\delta}{8192} b_a(j) = \frac{28}{64a^3} b_a(j) \end{aligned}$$

for $j \geq a/4$. This exceeds the possible negative part $25b_a(j)/(64a^3)$ of (35). At every other index that defect is already nonnegative. Hence

$$Q * R_{a'} \geq_{\text{st}} P * R_a. \quad (37)$$

B.3 From dominance to an exact causal tree

Maintain a virtual reserve Z_t and excess $D_t \geq 0$, with $J_t = Z_t + D_t$. Initially take $Z_0 \sim R_{a_n}$ and $D_0 = 0$. Conditional on each output history with $r = n - t$ steps remaining, maintain $Z_t \sim R_{a_r}$.

Draw fresh $Y \sim Q$ and set $S = Z_t + Y$. Use (37) to couple

$$S \sim Q * R_{a_r}, \quad W \sim P * R_{a_{r-1}}, \quad W \leq S.$$

This is an explicit quantile kernel: given $S = s$, choose a uniform point in the CDF interval of s under the first law and evaluate the second law's quantile there. Given $W = w$, select $X = x$ with probability

$$\frac{P(x)b_{a_{r-1}}(w-x)}{(P * R_{a_{r-1}})(w)}.$$

Set $Z_{t+1} = W - X$ and $D_{t+1} = D_t + S - W$. The pair (X, Z_{t+1}) , conditional on the preceding output history, has product law $P \otimes R_{a_{r-1}}$. Thus the outputs are exactly iid, and

$$J_{t+1} = J_t + Y - X \geq 0.$$

The accumulated excess is retained rather than discarded, so this is exact zero repair. All finite-horizon first moments are finite since $J_t \leq J_0 + 4t$.

B.4 Cost

For $a \geq 8$, write $x = 512/a^4 \leq 1/8$. Then

$$\left(a + \frac{512}{a^3}\right)^4 - a^4 = 512(4 + 6x + 4x^2 + x^3) \leq 2465.$$

Therefore $a_n^4 \leq 4096 + 2465n \leq 6561n$ for $n \geq 1$. The initial mean $4a_n$ proves (10). The virtual reserve at remaining depth zero has mean 32, which is allowed because no terminal zero-reserve condition is imposed.

C Why permutation-invariant infinite trees fail

Set $u = 1/64$, $p_0 = P(0) = 1/8$, and

$$\rho = \frac{q(u)}{p_0} = \frac{122725639}{134217728} < 1, \quad \gamma = \frac{\log \rho}{\log u} > 0.$$

Nonnegativity of the $x = 0$ term of (3) implies

$$F_{h0}(u) \leq \rho F_h(u).$$

Suppose permuting histories leaves their laws unchanged. Move all $N_0(h)$ zeros to the end of a history and iterate this inequality, bounding the preceding PGF by one. Then

$$F_h(u) \leq \rho^{N_0(h)}. \quad (38)$$

For finite means Jensen gives $u^{m_h} \leq F_h(u)$, hence $m_h \geq \gamma N_0(h)$. Averaging at depth n and using (12) gives (21).

For an infinite tree, infinite means require a different argument. Choose $u < v < 1$ and put $\theta = \log v / \log u \in (0, 1)$. Concavity implies

$$F_h(v) = \mathbb{E}[(u^J)^\theta] \leq F_h(u)^\theta \leq v^{\gamma N_0(h)}.$$

On the other hand, iterating the exact equation gives

$$q(v)^n F_\emptyset(v) = \mathbb{E}_{P^{\otimes n}}[v^{S(h)} F_h(v)].$$

The upper bound factorizes, yielding

$$F_\emptyset(v) \leq K(v)^n, \quad K(v) = \frac{p(v) + p_0(v^\gamma - 1)}{q(v)}.$$

Since $K(1) = 1$ and $K'(1) = p_0\gamma > 0$, there is a fixed $v < 1$ sufficiently close to one with $K(v) < 1$. Sending $n \rightarrow \infty$ forces $F_\emptyset(v) = 0$, impossible for a proper probability law on $\mathbb{Z}_{\geq 0}$. This proves Proposition 3.2.

For the narrower class $B_h = B_{|h|, S(h)}$, one obtains a stronger finite-horizon bound. A sum s can be attained in $k = \lceil s/4 \rceil$ steps, followed by $n - k$ zeros. Thus $m_{n,s} \geq \gamma(n - \lceil s/4 \rceil)$. The distribution P modulo four is uniform; consequently S_n modulo four is uniform for $n \geq 1$ and

$$\mathbb{E}[\lceil S_n/4 \rceil] = n/2 + 3/8.$$

Therefore $\mathbb{E}B_\emptyset \geq \gamma(n/2 - 3/8)$ in this narrower class. Neither estimate restricts arbitrary order-dependent trees.

D What a critical reference profile does and does not cancel

For $\lambda > 0$ define

$$\pi_\lambda(j) = \frac{\lambda}{(j + \lambda)(j + \lambda + 1)}, \quad j \geq 0.$$

Telescoping proves normalization and the exact survival formula

$$\sum_{k \geq j} \pi_\lambda(k) = \frac{\lambda}{j + \lambda}.$$

For the illustrative tilt $z_* = 1/16$, let $\ell_x = z_*^x/p(z_*)$. Then $\sum_x P(x)\ell_x = 1$. Thus a child tail coefficient $\lambda\ell_x$ preserves the leading j^{-2} coefficient after P -averaging. This is a correct leading-order observation, but the next order already matters.

Put $m_* = \sum_x P(x)\ell_x x = 548/4369$ and $M_2 = \sum_x P(x)\ell_x^2 = 524296/83521$. For fixed λ ,

$$\pi_\lambda(j) = \lambda j^{-2} - \lambda(2\lambda + 1)j^{-3} + O_\lambda(j^{-4}).$$

Replacing j by $j - x$ and averaging gives

$$(Q * \pi_\lambda)(j) - \sum_x P(x)\pi_{\lambda\ell_x}(j - x) = \frac{2\lambda(2 - m_*) + 2\lambda^2(M_2 - 1)}{j^3} + O_\lambda(j^{-4}). \quad (39)$$

The coefficient is positive. The varying child scales therefore introduce a j^{-3} mass correction before the j^{-6} mass defect arising from the fourth difference of one fixed smooth profile. Fixed- λ estimates also provide no uniform control as λ ranges over histories.

If the tail coefficient of B_h were Λ_h , (20) would prescribe

$$\tau(H_{h,i}) = \Lambda_h \sum_{x=0}^i P(x)(\ell_x - 1), \quad 0 \leq i \leq 3.$$

This only fixes four asymptotic coefficients. It neither proves nonnegativity nor permits the whole law to forget order. The profile is a diagnostic example; it is not a hypothesis of Problem 2.5 or a required route to its solution.

Provenance and scope of the references

The motivating source [15] is an author-supplied manuscript, used for the distinction between realization contracts. The finite contact facts are taken from the supplied project calculations [5]; their large rational certificates are not reproduced in this note. The all-horizon estimates and structural reserve lemmas have proofs above. External results are attributed where used, and no priority claim is made for applying established mechanisms. The literature map is targeted rather than exhaustive. There is no mathematical transfer here to SAT witness extraction, physical ontology, or a computational substrate model.

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